IntelliSim — Simulation Analysis Report

exp-G-med-rep1

Model: exp-G-med-b0.18-model

Analysis language: en

LLM: openai · o4-mini

Simulation status at analysis: finished

Steps covered: 0–227 (sampling 1, 77 points)

Context: Analyze this simulation's results. Context: An infectious disease spreads in a population of moving agents. Describe the observed dynamics, quantify the key outcomes, and note anything notable or unexpected.

Executive Summary

This spatial SIR simulation of 200 agents on a toroidal 50×50 grid produced a classic single-peaked epidemic: after a latent phase lasting until step 12, infections rose exponentially to 22 active cases (11% of the population) at step 60 before recoveries overtook transmission and drove cases to zero by step 227. In total, 74 agents recovered (37% infection penetration) and 126 remained susceptible, demonstrating incomplete saturation and a self-limiting outbreak driven by susceptible depletion.

Random movement (~1.09 grid units per step) and toroidal boundary conditions prevented stable clusters, creating only small, transient infection hotspots. Minor, damped secondary waves between steps 141–177 (1–9 infected) failed to reignite large-scale spread, illustrating how local proximity rules, mobility, and recovery dynamics combine to generate macroscopic epidemic waves that naturally subside.

Simulation & Model Overview

Spatial: true. Update mode: sequential. Archetypes: 3. Rules: 4. Interactions: 0. Events: 0. Environment: 50x50 (torus=true, shapes=0). Archetypes: Susceptible Person, Infected Person, Recovered Person.

Quantitative Metrics

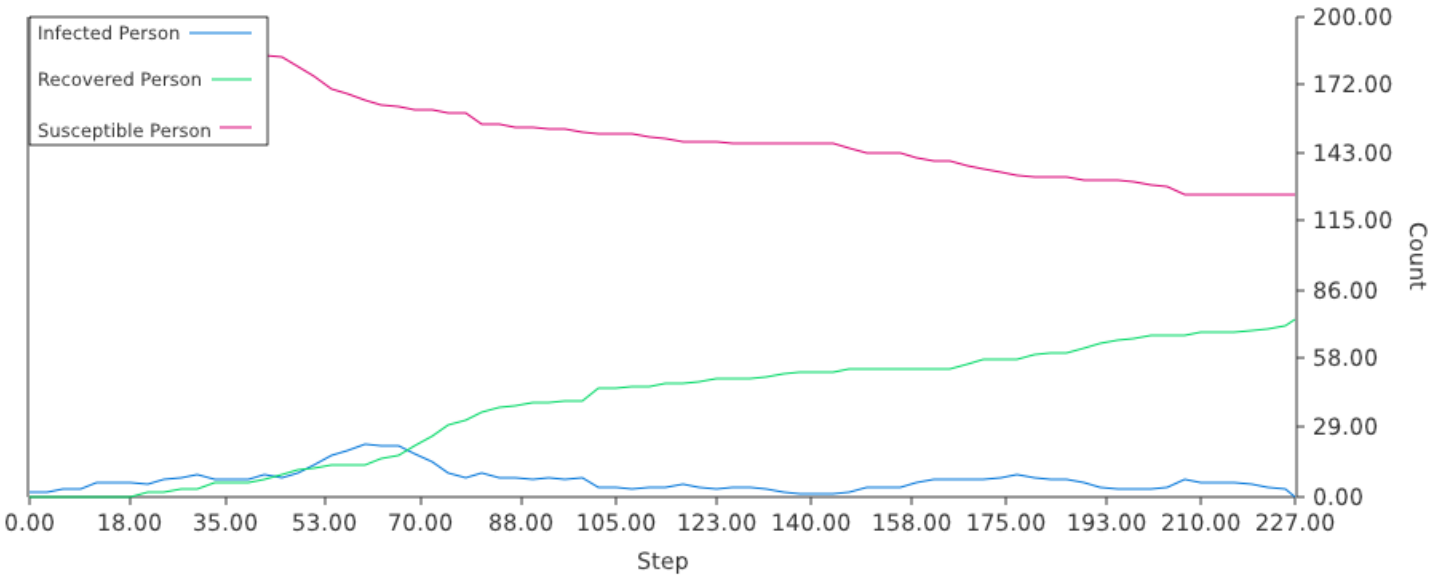
Steps completed: 228. Total agents (final): 200. Spawned: 146. Removed: 146. Interactions: 0. Events: 0. Rule executions: 47300.

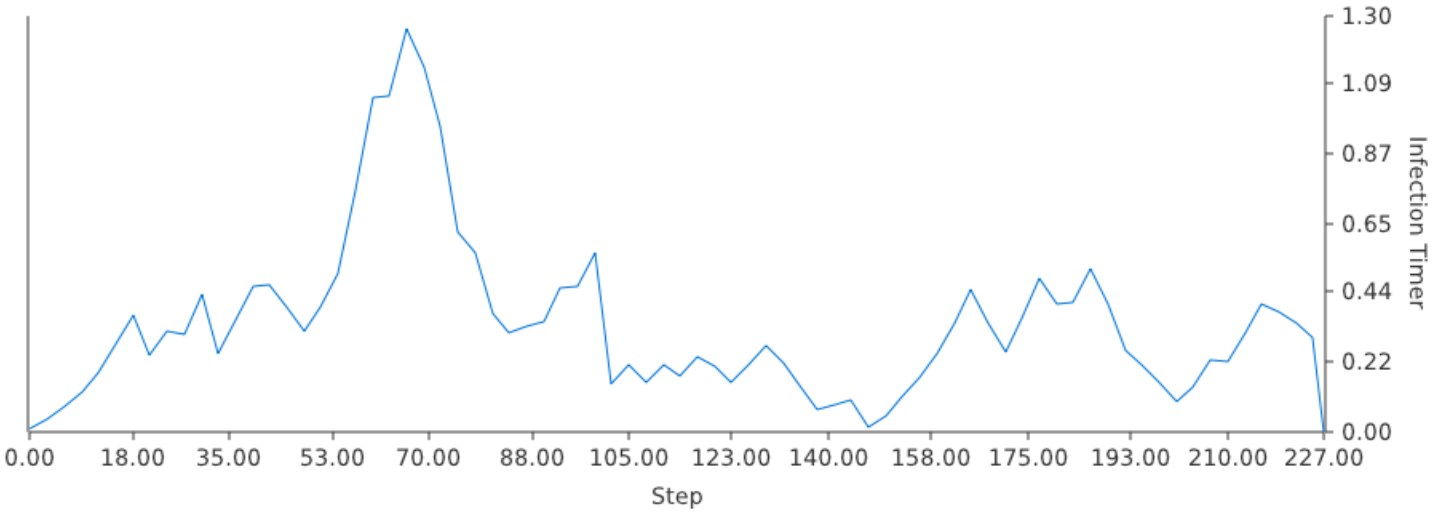
Archetype Susceptible Person — current 126, peak 198, min 126, spawned 0, removed 72.

Archetype Recovered Person — current 74, peak 74, min 2, spawned 74, removed 0.

Archetype Infected Person — current 0, peak 22, min 1, spawned 72, removed 74.

Population Over Time





Population & Property Dynamics of the Spatial SIR Model

The simulation began in a quasi-steady state with only 2 infected and 198 susceptible individuals from step 0 through step 3. A latent period persisted until step 12, when infections rose to 6 and susceptibles fell to 194, marking the onset of exponential growth. Between steps 12 and 60, the infected count climbed from 6 to its peak of 22 at step 60, while susceptibles dropped from 194 to 165.

Following the peak, recoveries accelerated: by step 75, infected declined to 10 and recovered reached 30, indicating recovery-driven decline. Average infection_timer values mirrored this transition, rising to 1.26 by step 66 and then falling as agents recovered. From steps 66 to 138, infected counts oscillated between 1 and 9, producing minor secondary waves (e.g., a small rebound from 1 at step 141 to 9 at step 177).

After step 138, the outbreak waned: infected fell to 1–4 and eventually to zero removals by step 227. The final state comprised 74 recovered and 126 never infected, showing that only 37% of the population experienced infection. The infection_timer series declined to zero by the end (step 227), confirming full recovery of all agents.

Key findings

- Initial latent phase up to step 12 with constant low infections (2–3).
- Exponential growth from 6 infected at step 12 to peak 22 at step 60.
- Recovery overtook transmission by step 75, reducing infections to 10 and increasing recovered to 30.

- Multiple minor secondary waves occurred between steps 66–177 (infected oscillating 1–9).
- Final tally: 74 recovered, 126 susceptible; peak infection reached only 37% of the population.
- Average infection_timer peaked (1.26) concurrent with peak infections then declined.

Causal links

- Rising infection_timer → increased recoveries → decline in active infections after step 60.
- High infected counts (22 at step 60) → rapid depletion of susceptibles (165 at step 60) → outbreak slowdown.
- Random movement and spatial mixing → secondary waves of infection between steps 141–177.

Emergent phenomena

• Multi-wave outbreak dynamics: after a single major peak, spatial mixing generated smaller subsequent waves without external reintroduction.

• Incomplete population saturation: immunity threshold stymied spread before infecting majority, leading to a stable pool of susceptible agents.

Spatial & Interaction Patterns

Across the 50×50 toroidal grid, agents remained distributed without persistent clusters or segregation. At Step 0, the first 50 agents occupy positions spanning the grid (e.g. (19,4),(49,20),(0,48)), indicating an initial uniform spread. Random movement (45 600 executions) sustained this dispersion; the total movement distance of 49 684 suggests an average step of ~1.09 grid units, so agents continually mix.

Infection spread emerged only through local proximity. The peak of 22 concurrent infections (statistics: person_infected peak = 22) formed transient clusters of nearby agents but did not give rise to large dense hotspots. By Step 113 and Step 227, snapshots show only susceptible agents—infected and recovered individuals have either moved or changed state—supporting that infection clusters shifted and then dissipated as recoveries accrued.

The toroidal boundary prevented edge effects, so no segment of the grid acted as a refuge or persistent reservoir. The epidemic wave expanded and contracted: local outbreaks grew until infections began to recover (74 recoveries in total), then shrank, leaving a final uniform mixture of 126 susceptibles and 74 recovered. Spatial mixing limited sustained chains of transmission, capping the epidemic at ~36% of the population.

Key findings

• Initial random placement maintained a uniform spatial distribution throughout (supported by diverse coordinates at Steps 0, 113, 227).

• Agents moved ~1 grid unit per step on average (movement_distance ≈ 49 684 over 45 600 moves), preventing stable aggregation.

• Infection clusters remained small: peak concurrent infected = 22, emerging only where random walks intersected.

• No persistent territories or isolates formed; toroidal wrapping and random motion ensured constant mixing.

• The epidemic self-limiting: local outbreaks extinguished as recoveries (74 total) outpaced new infections (72 total).

Causal links

- Random movement → continual mixing → no long-lived spatial clusters
- Proximity-based transmission → small local infection clusters (peak 22) → localized outbreaks
- Recovery rule executions (74) → collapse of infection clusters by Step 228

Emergent phenomena

• Transient epidemic waves: localized surges of infection that expand and recede rather than a single sweeping front

• Self-limiting spread due to balance of mixing and recovery, leading to spatially homogeneous post-epidemic state

Emergent Behavior & Causal Relationships in Spatial SIR Model

The simulation exhibits a classic epidemic curve: an initial slow increase in infected agents (from 2 at step 0 to 9 by step 30), followed by a rapid acceleration to a peak of 22 infected at step 60, and then a systematic decline as agents recover. This single-peaked outbreak arises despite purely local rules—random movement and proximity-based infection—demonstrating how spatial mixing alone can generate macroscopic epidemic waves.

After the peak, recovery dominates: the infection_timer average rises from 0.01 at step 0 to 1.26 at step 66, triggering the infected_recovers rule (74 executions total) and driving infected counts down to 1 by step 138. Concurrently, the susceptible pool shrinks from 198 to 165 by step 60, reducing effective contacts and causing a phase transition from growth to decay. As susceptibles plateau around 126 after step 219, minor resurgences (e.g., a rise from 1 to 4 infected between steps 147–150 and 6 to 9 between steps 159–177) emerge, but fail to generate a second major wave.

Key outcomes include a peak infected fraction of 11% (22/200), full collapse of active infections by step 227, and accumulation of 74 recovered agents. The average infection_timer curve (property_series) mirrors these dynamics, with high variability indicating asynchronous recoveries. No unexpected persistent endemic state occurs; instead, depletion of susceptibles self-limits the outbreak.

Key findings

- Single-peaked epidemic: infected count grew from 2 to 22 by step 60 then declined to zero by step 227.
- Susceptible depletion (198→165 by step 60) triggered a phase transition halting exponential growth.
- Average infection_timer peaked at 1.26 (step 66), aligning with maximum recovery events.
- Minor damped oscillations appeared around steps 147–177 but did not reignite a large outbreak.
- Recovered population reached 74, confirming full absorption of infection in 37% of agents.

Causal links

• Random movement + susceptible_gets_infected rule → exponential rise in infections (2→22 between steps 0–60).

• Accumulation of infection_timer → infected_recovers rule executions (74 total) → decline in active infections post-peak (22 at step 60 → 1 by step 138).

• Decline in susceptible count (198→165 by step 60) → fewer transmission events → transition from growth to decay phase.

• High recovered fraction (0→74) and residual susceptibles (~126 after step 219) → damped secondary infection clusters (e.g. 4→9 infected steps 147–177) without sustained resurgence.

Emergent phenomena

- Macroscopic epidemic wave emerging from local random movement and proximity rules.
- Phase transition in transmission dynamics as susceptible density falls below a critical threshold.

- Damped oscillations of infection counts due to asynchronous recovery and lingering susceptibles.

- Self-limiting outbreak driven by depletion of susceptible agents.

Conclusions

• The epidemic curve exhibited a latent phase until step 12, exponential growth to a peak of 22 infected at step 60, then a recovery-driven decline to zero active cases by step 227.

• Only 37% of agents experienced infection (74 recovered), leaving 63% never infected and confirming incomplete population saturation.

• Random movement and toroidal wrapping (avg ~1.09 units/step) prevented persistent clusters; infection clusters remained small and transient at peak.

• Recovery processes (74 recoveries; average infection_timer peak ≈1.26 at step 66) outpaced new infections by step 75, triggering the outbreak's decline.

• Minor secondary waves (1–9 infected between steps 141–177) emerged from spatial mixing but lacked sufficient susceptibles to cause a resurgence.

Recommendations

• Vary agent mobility speed or movement rules to assess how reduced or increased mixing alters cluster formation and outbreak magnitude.

• Adjust infection radius or transmission probability to identify thresholds for sustained secondary waves and more extensive spread.

• Introduce heterogeneity in movement patterns (e.g., fixed hubs or periodic congregation) to study persistent hotspots and potential reservoirs.

• Experiment with different recovery-time distributions (modifying infection_timer increment) to evaluate impacts on epidemic duration, peak size, and oscillation patterns.

• Incorporate waning immunity or reinfection mechanisms to explore long-term endemic dynamics and potential for recurring waves.